TUNI Luottamuksellinen - Confidential

Tampereen yliopisto
Tampere University

Laboratory of Biosystem Dynamics

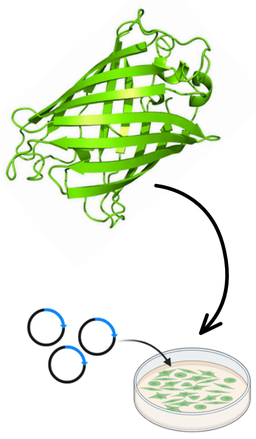
Funded by:

GXA: A tool for characterizing bacterial gene expression dynamics measured by time-lapse spectrophotometry

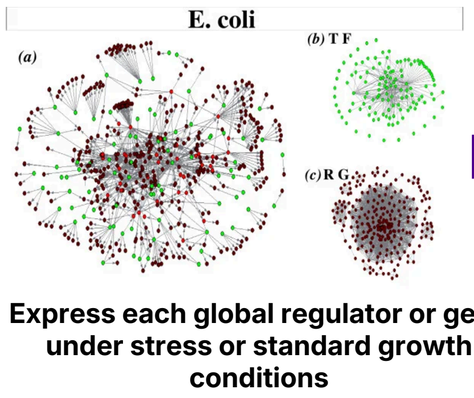
Rohan Ghosh Dastidar[1] and Andre S. Ribeiro[2]
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[2]Laboratory of Biosystem Dynamics, Faculty of Medicine and Health Technology, Tampere University, Finland.
E-mail: rgdastidar2069@kgpian.iitkgp.ac.in; andre.sanchesribeiro@tuni.fi

Introduction

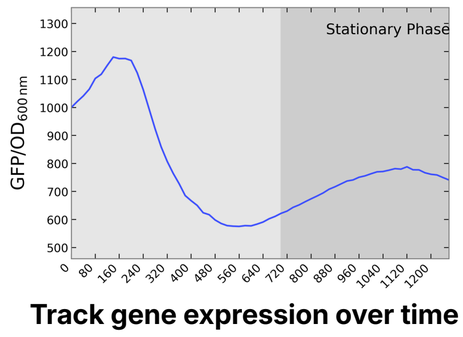
Tracking protein numbers over time reveals the dynamics of genes and how the underlying gene network controls them



GFP/YFP tagged proteins



Express each global regulator or gene under stress or standard growth conditions



Track gene expression over time

BUT No way to automatically extract quantitative features from the data

Or to easily compare gene expression across different strains/conditions/ perturbations

We developed a tool to use Gene Expression Time Series data to classify

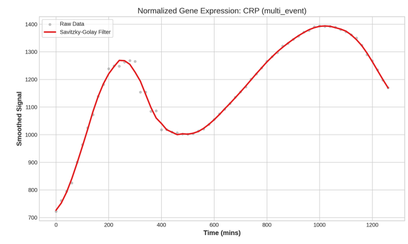
SHAPE
- monotonic
- single peak/trough
- biphasic
- oscillatory

FEATURES
- Rise/fall time
- Fold change
- Peak/Trough
- Stationary phase
- Fold change
- Peak/Trough

Time	CRP
0	721
20	761
40	794
60	825
80	900
100	963
120	1025
140	1071
160	1138
180	1181.25

Data

Gene eXpression Analyzer



Shape classification

```
CRP:
  baseline: 726.1564001690152
  baseline_method: initial_value
  category: biphasic
  metadata:
    num_peaks: 2
    metrics:
      transition_delay: 780.0
      net_amplitude_ratio: 4.4102031980091665
      phase_duration_ratio: 1.7391304347826086
      composite_symmetry_score: 0.401
      phase_dominance_profile: secondary_phase_dominant
      inter_phase_clearance_pct: 178.8
      amplitude_symmetry_component: 0.227
      temporal_symmetry_component: 0.575
```

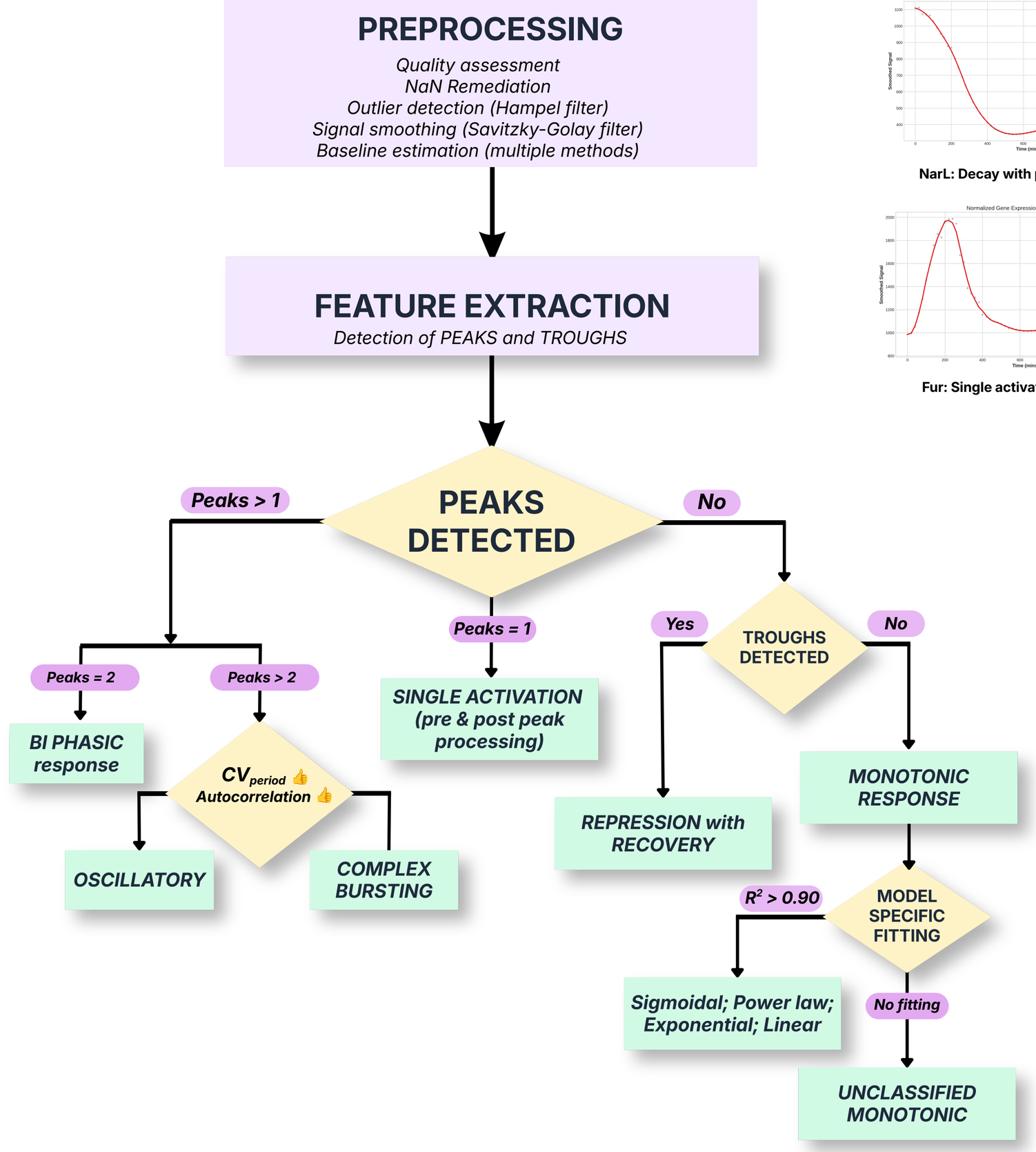
Details

Methods/Algorithm

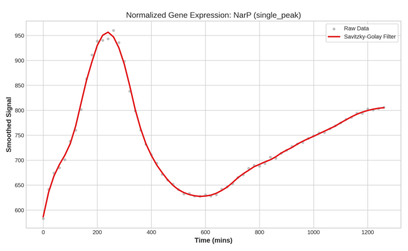
INPUT
Gene expression time series (.csv)

PREPROCESSING
Quality assessment
NaN Remediation
Outlier detection (Hampel filter)
Signal smoothing (Savitzky-Golay filter)
Baseline estimation (multiple methods)

FEATURE EXTRACTION
Detection of PEAKS and TROUGHS

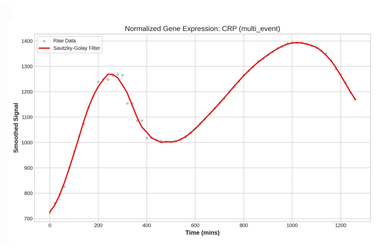


Results & Conclusion



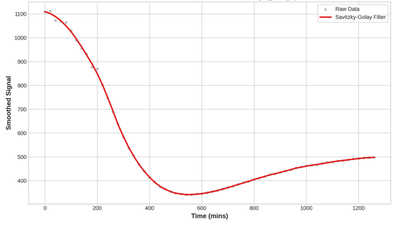

```
NarP:
  baseline: 586.569933362517
  baseline_method: initial_value
  category: single_peak
  metadata:
    pre_peak_behaviour:
      category: clean_monotonic_rise
    peak_behaviour:
      peak_time: 240.0
      absolute_amplitude: 956.5999407110622
      net_rise: 370.03000734854527
      fold_change: 1.6308369834562526
      rise_time: 120.0
      fall_time: 120.0
      fwhm_width: 240.0
      asymmetry_ratio: 1.0
      total_yield_auc: 211803.4785301755
    post_peak_behaviour:
      category: decrease_and_recovery
      baseline_minima: 627.3651419248556
      percentage_recovery: 48.07748816619737
```

NarP: Single activation with partial recovery



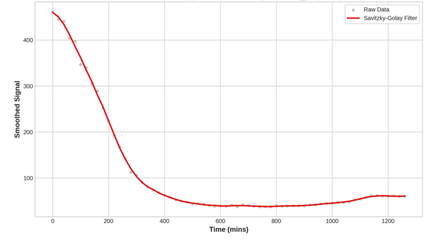

```
CRP:
  baseline: 726.156
  baseline_method: initial_value
  category: biphasic
  metadata:
    num_peaks: 2
    metrics:
      transition_delay: 780.000
      net_amplitude_ratio: 4.410
      phase_duration_ratio: 1.739
      composite_symmetry_score: 0.401
      phase_dominance_profile: secondary_phase
      inter_phase_clearance_pct: 178.800
      amplitude_symmetry_component: 0.227
      temporal_symmetry_component: 0.575
```

CRP: Bi-phasic response



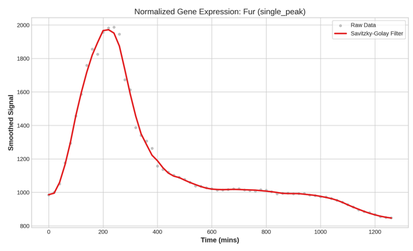

```
NarL:
  baseline: 1109.4560042738167
  baseline_method: initial_value
  category: single_trough
  metadata:
    pre_trough_behaviour:
      category: immediate_dip
      decline_rate: 1.3713427231877489
      time_to_minimum: 560.0
    trough_behaviour:
      trough_time: 560.0
      trough_value: 341.5040792886773
      baseline_value: 1109.4560042738167
      net_drop: 767.9519249851394
      fold_change_down: 3.248734265736196
      depth_as_pct_of_baseline: 69.21878127901022
    post_trough_behaviour:
      category: partial_recovery
      recovery_time: 700.0
      recovery_amount: 156.390921852308
      recovery_pct: 20.364676064237857
      final_value: 497.8950011409853
      stabilization_point: 494.6690641200386
```

NarL: Decay with partial recovery



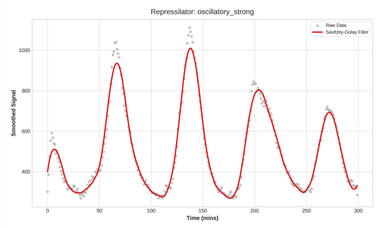

```
MG1655:
  baseline: 459.832
  baseline_method: initial_value
  category: single_trough
  metadata:
    pre_trough_behaviour:
      category: gradual_decline
      decline_rate: 0.555
      time_to_minimum: 760.000
    trough_behaviour:
      trough_time: 760.000
      trough_value: 37.800
      baseline_value: 459.832
      net_drop: 422.032
      fold_change_down: 12.165
      depth_as_pct_of_baseline: 91.780
    post_trough_behaviour:
      category: partial_recovery
      recovery_time: 500.000
      recovery_amount: 22.894
      recovery_pct: 5.425
      final_value: 60.694
      stabilization_point: 60.770
```

MG1655 (Control): Monotonic decrease



Fur: Single activation with decay

```
Fur:
  baseline: 986.346
  baseline_method: initial_value
  category: single_peak
  metadata:
    pre_peak_behaviour:
      category: clean_monotonic_rise
    peak_behaviour:
      peak_time: 220.000
      absolute_amplitude: 1971.725
      net_rise: 985.379
      fold_change: 1.999
      rise_time: 120.000
      fall_time: 100.000
      fwhm_width: 220.000
      asymmetry_ratio: 0.833
      total_yield_auc: 391461.199
    post_peak_behaviour:
      category: continued_decay
      total_drop_percent: 57.065
      final_value: 846.555
```




```
Protein_Signal:
  baseline: 461.620
  baseline_method: initial_value
  category: oscillatory_strong
  metadata:
    num_peaks: 4
    metrics:
      success: True
      error_msg: None
      periodicity:
        mean_period: 68.000
        period_regularity_cv: 0.032
        cycles_per_hour: 0.882
      amplitude_profile:
        max_peak_intensity: 929.354
        min_peak_intensity: 650.794
        mean_peak_intensity: 805.761
        peak_intensity_cv: 0.128
        mean_trough_floor: -280.971
        mean_pulse_height: 1086.733
      attenuation:
        damping_gradient_per_pulse: -0.089
        terminal_attenuation_ratio: 0.759
        envelope_classification: damped_oscillations
    confidence_score: 0.900
```

Repressilator: Oscillatory

Discussion

- Manually characterizing the response of gene circuits is very difficult in synthetic biology.
- GXA will automate this process and enable us to have direct, unbiased comparisons of bacterial gene expression across diverse host strains and growth media.
- This tool can also be paired with stochastic simulation pipelines which will allow users to quantify the effect of perturbation of different parameters in their models

Future Directions

We plan to improve this tool by adding more motifs such as switching, transcriptional bursts, and other slightly chaotic models

References

Dash, S., Jagadeesan, R., Baptista, I. S. C., Chauhan, V., Kandavalli, V., Oliveira, S. M. D., & Ribeiro, A. S. (2024). A library of reporters of the global regulators of gene expression in Escherichia coli. mSystems, 9(6), e00065-24.

Acknowledgement

The author's research was generously funded by the Platinum Jubilee Outbound Internship Scheme from the Indian Institute of Technology Kharagpur. We also thank the Faculty of Medicine and Health Technology of Tampere university for providing the resources for this project. The illustrations were created with BioRender.com.

17th Annual COMBINE Conference 2026

GXA: A software for analyzing Gene Expression time-series data measured by time-lapse spectrophotometry

Rohan Ghosh Dastidar

Final year, Dual Degree, Department of Chemical Engineering

Indian Institute of Technology Kharagpur, India

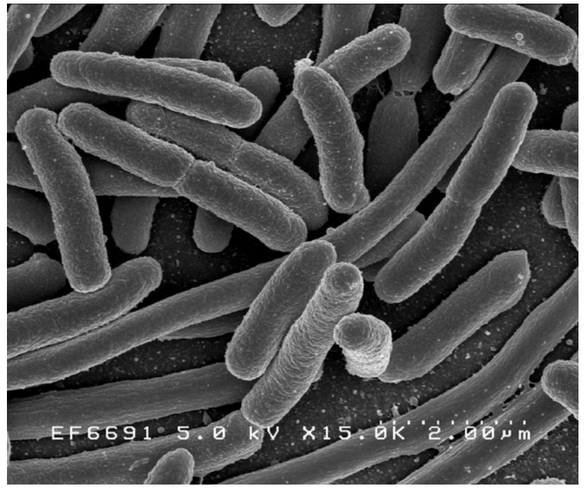
Co-author/Supervisor:

Prof. Andre S. Ribeiro

Professor, Faculty of Medicine & Health Technology, Tampere University

Principal Investigator, Laboratory of Biosystem Dynamics

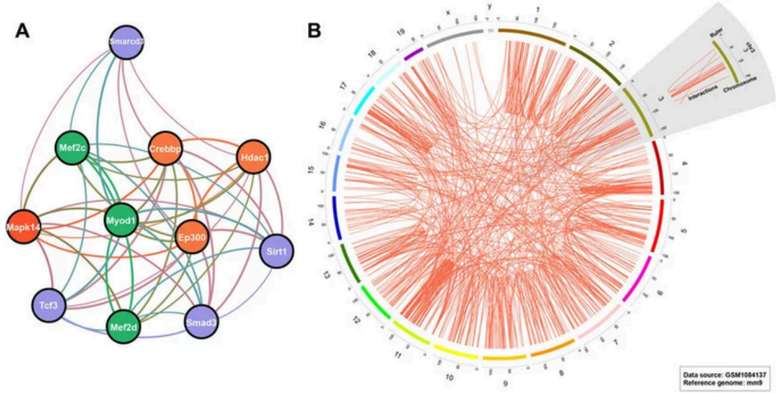
INTRODUCTION & BACKGROUND



E. coli cells under a microscope
Source: National Geographic

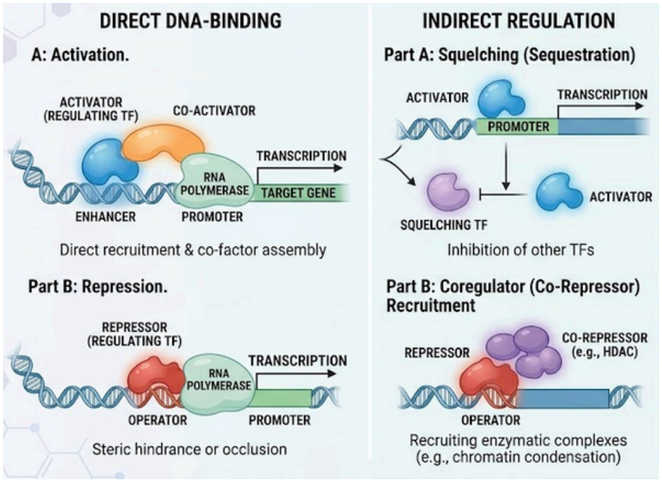
How do cells respond to external signals??

Through GENE REGULATORY NETWORKS



Different visualizations of gene regulatory networks.
Source: https://doi.org/10.1016/j.csbj.2023.02.028

Methods of gene regulation

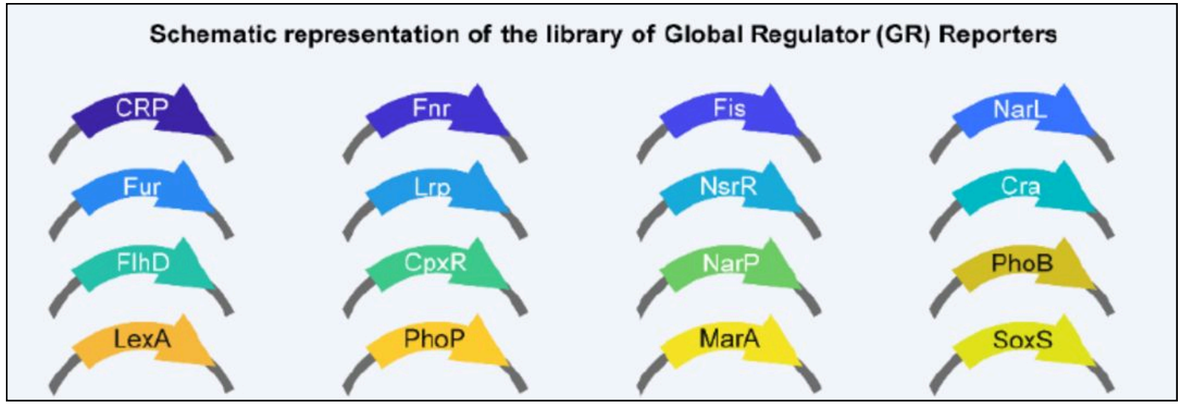


Overview of molecular mechanisms of gene regulation, Source: AI generated, Gemini (Google)

Now, E. coli. has around 4000 genes

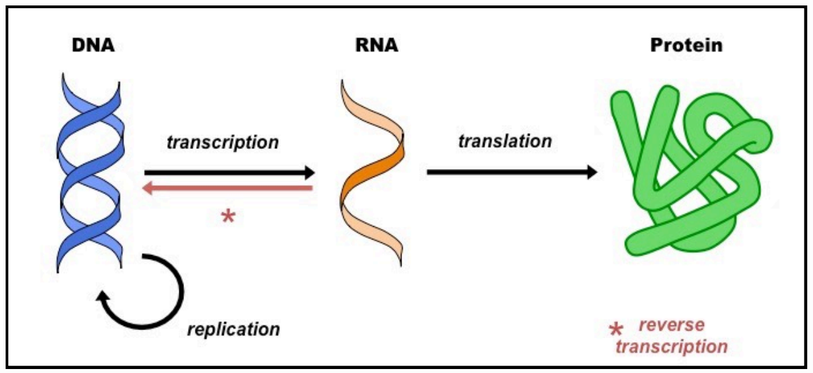
But, all of them don't behave independently!

Nearly 30% of genes are controlled by 22 specific genes, called **GLOBAL REGULATORS**



Schematic representation of E. coli. global regulators,
Source: doi: 10.1128/msystems.00065-24

LITERATURE REVIEW



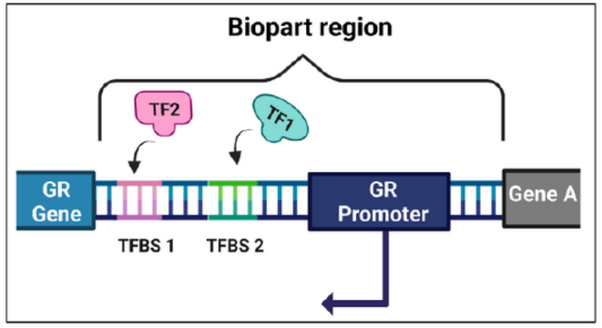
Central Dogma of molecular biology,
Source: https://personal.cityu.edu.hk/liangdai/post/central-dogma-translation-transcription/

AMERICAN SOCIETY FOR MICROBIOLOGY | mSystems
doi: 10.1128/msystems.00065-24.
Check for updates
| Systems Biology | Research Article

A library of reporters of the global regulators of gene expression in *Escherichia coli*

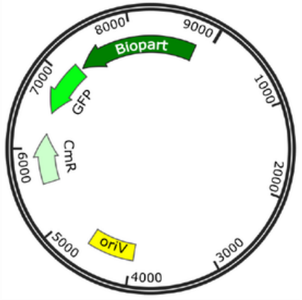
Suchintak Dash,[1] Rahul Jagadeesan,[1] Ines S. C. Baptista,[1] Vatsala Chauhan,[1] Vinodh Kandavalli,[2] Samuel M. D. Oliveira,[3] Andre S. Ribeiro[1]

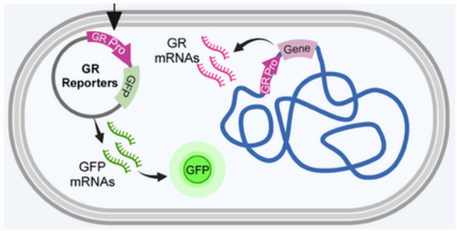
This paper builds a library of strains containing global regulators tagged with GFP



- **Expression is induced in exponential phase**
- **Protein numbers are tracked with fluorescence spectrophotometer**
- **Cell population is tracked with OD600**

Source: 10.1128/msystems.00065-24

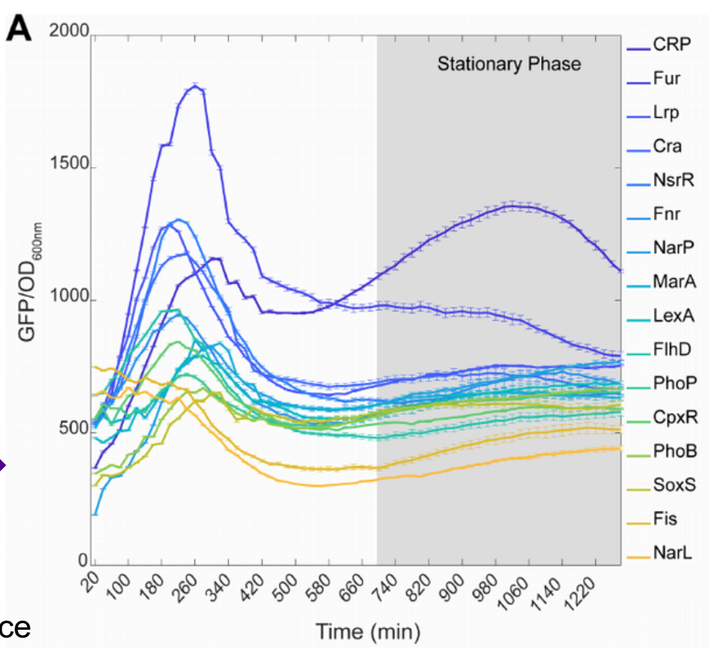








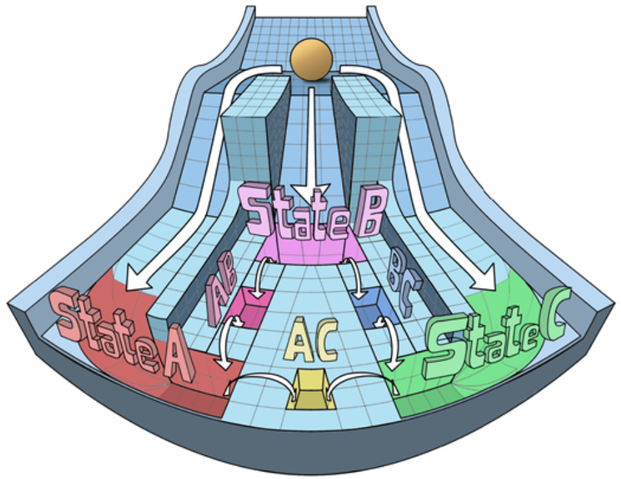
Spectrophotometer for OD600 and for fluorescence



Why do we need analyze gene expression data??

Cellular Decision-Making

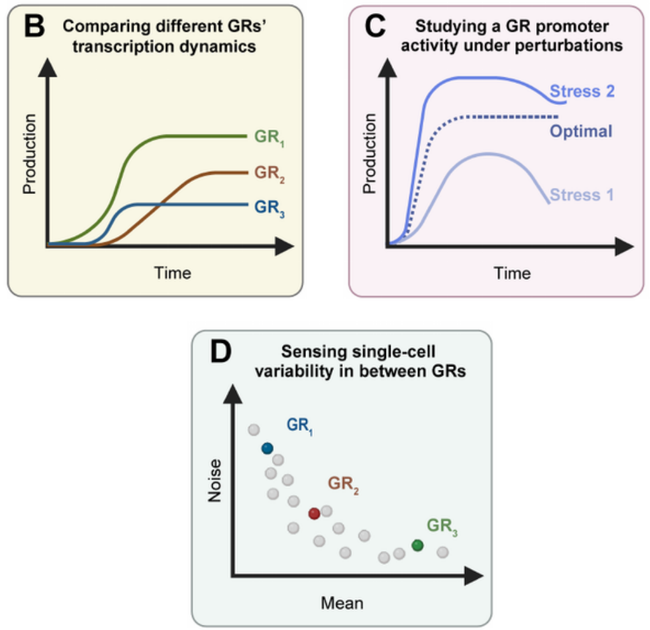
- Dynamic responses:
  - Genes do not just turn ON and OFF
  - Protein numbers exhibit complex patterns:
    - transient spikes,
    - steady states,
    - slow declines, or
    - rhythmic oscillations.
- Shape, speed, and duration of a gene's signal reveal how the cell senses environmental changes and executes survival strategies.



Cellular decision making: States vs Fates
Source: https://doi.org/10.1098/rsif.2020.0631

Comparing Strains & Conditions

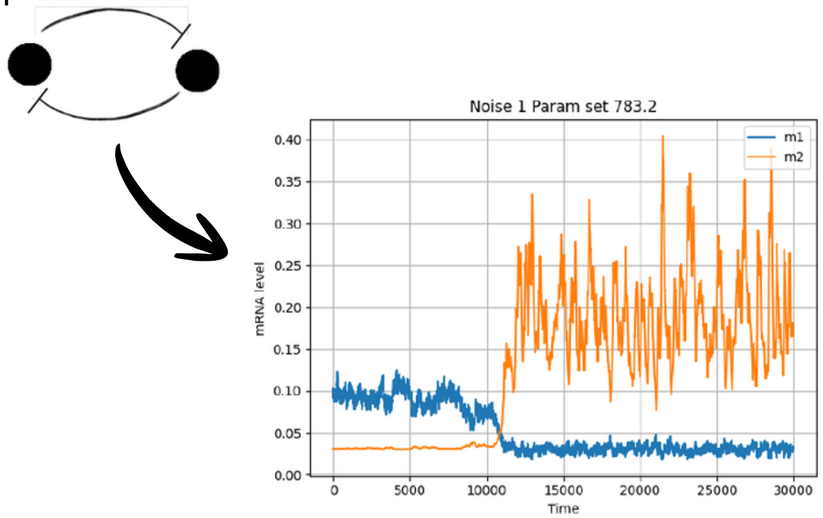
Compare cellular responses;
✓ Wild type vs mutated strain
✓ Different growth conditions
✓ Different stress signals



Comparison of gene response across conditions
Source: 10.1128/msystems.00065-24

Predictive Mathematical Modeling

Kinetic data (**peak timing, decay rates**) can be used to calibrate stochastic simulation models and predict system behavior under new parameters.



Stochastic simulations with variable parameter sets, Source: Original work

Why do we need analyze gene expression data??

HIGH DATA VOLUME

Extracting kinetic metrics manually is TEDIOUS, prone to ERROR, and hard to STANDARDIZE

Existing tools group similar overall shapes (clustering) rather than giving quantitative, interpretable metrics for single individual expression curves.

There MUST be a way to easily classify the response and obtain quantifiable metrics directly from the time-series data

This is where GXA comes in!

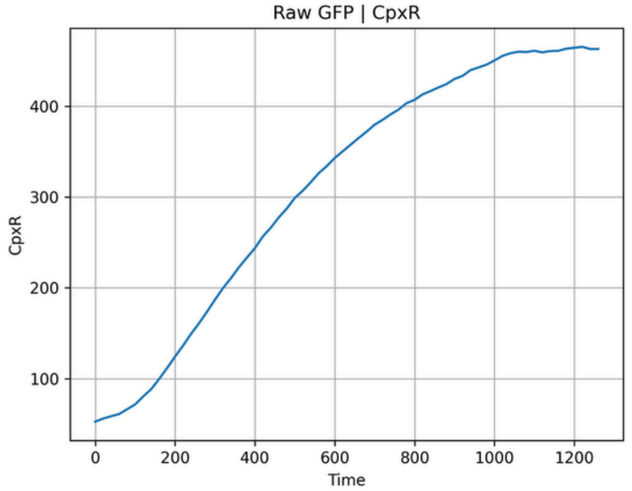
GXA
Gene eXpression Analyzer

Data Measurements

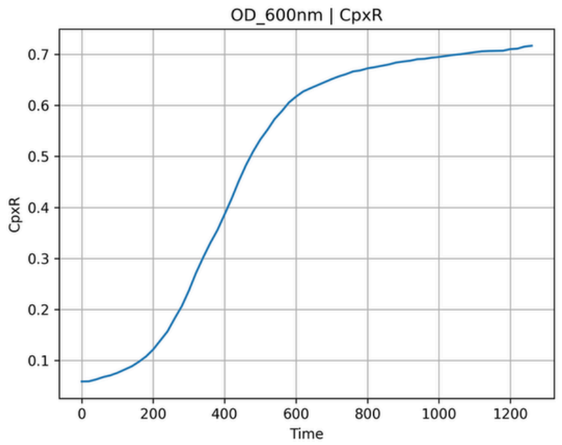
What's the input data?

- Fluorescence intensity (RFU)
- Optical Density (OD600)

Organism: Escherichia coli K-12
Substrain: MG1655
Standard growth conditions
Measurements every 20 mins, over 21 hours
Total: 63 data-points per samples
Replicates: 3 per sample

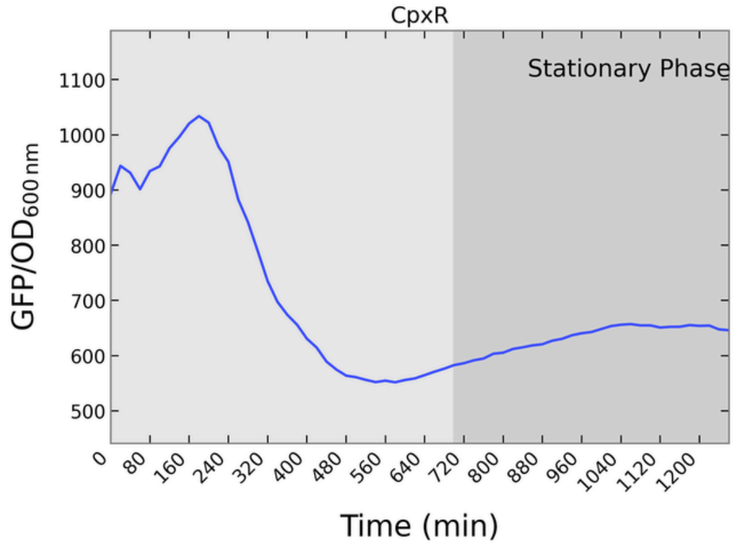


Raw fluorescence values



OD measurements

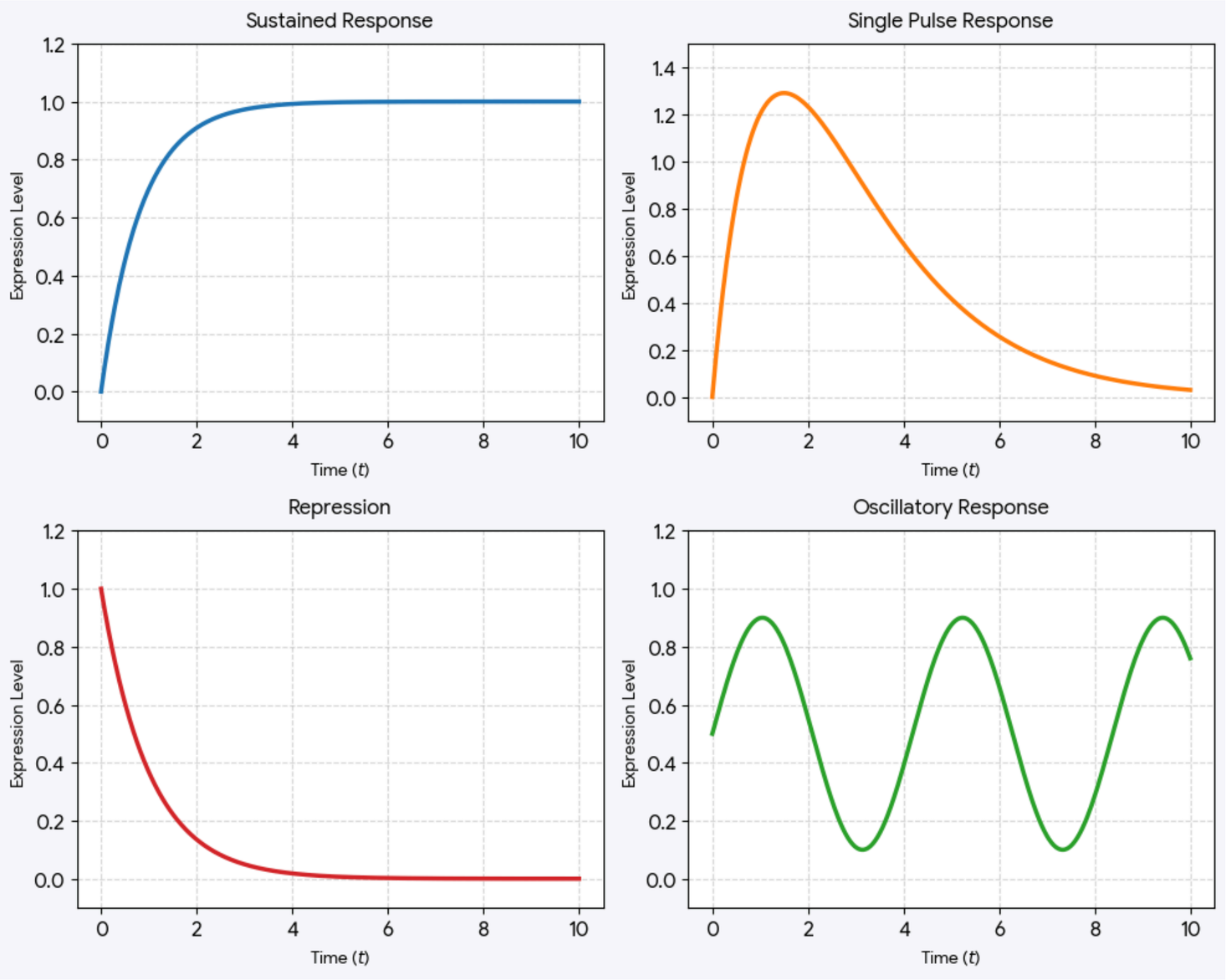
Gives the "average" response per cell >>



OD normalized fluorescence

Software Architecture

Response Phenotypes in Systems Biology



While these curves represent clean, theoretical shapes, our tool is specifically optimized to detect and separate them within noisy, mixed experimental signals.

Source: Original work

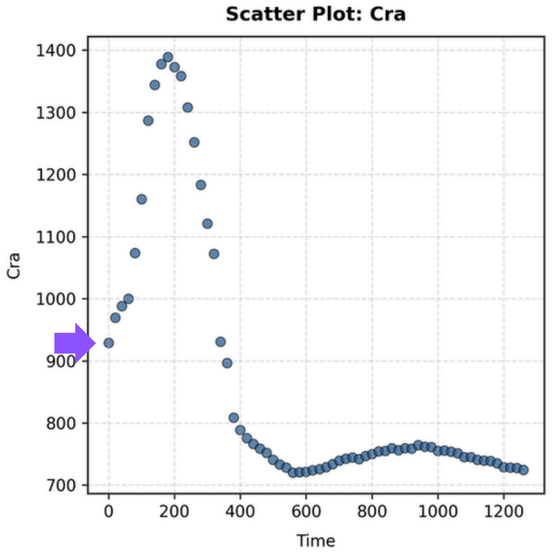
Software Architecture

INPUT VALIDATION

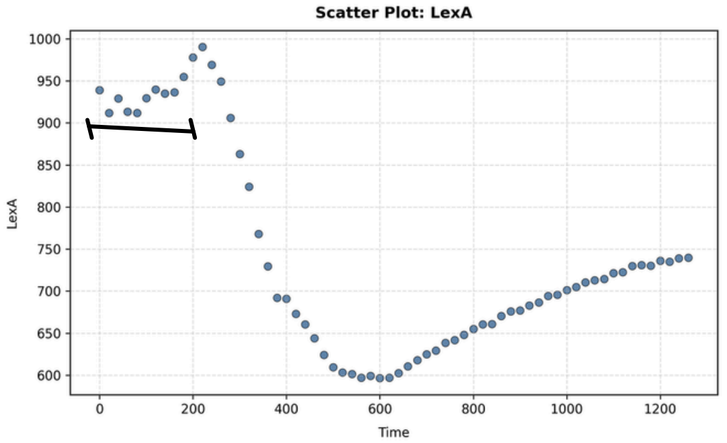
- Clean time-series data
- Missing Data > 50% (NaNs) values
- Missing data < 50%
- Valid RFU/OD

BASELINE ESTIMATION

- initial value
- initial window average
- initial median window



or

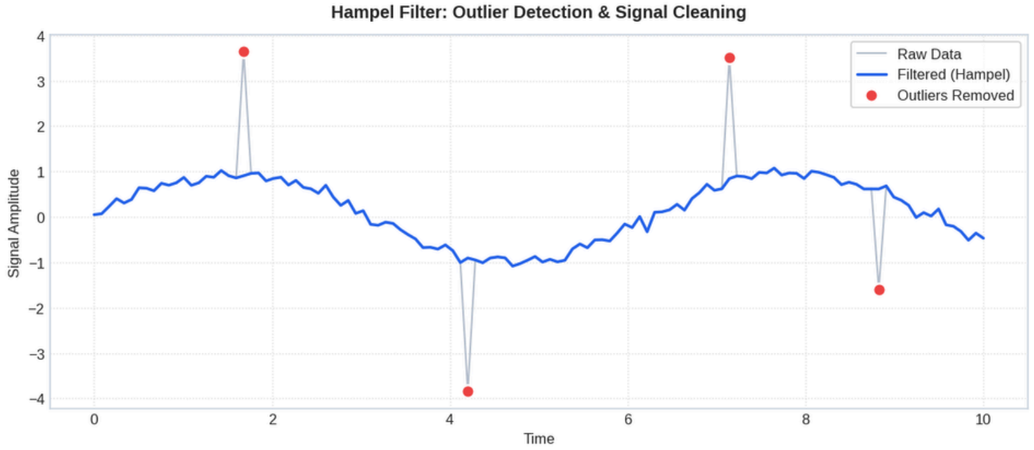


or User input

OUTLIER REMOVAL

Hampel filter

- Flags points exceeding 3 *MAD from rolling median
- Replaces outlier points using Piecewise Cubic Hermite Interpolation (PCHIP) to preserve smoothness

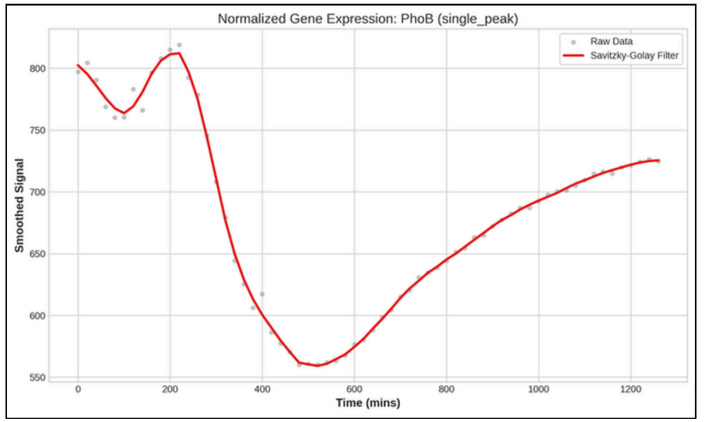


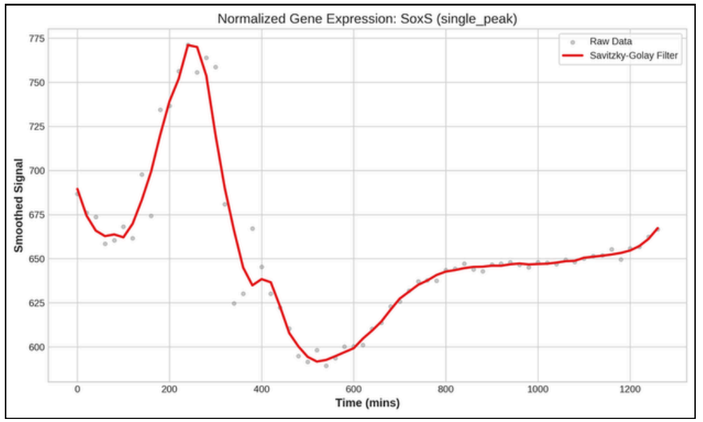
Source: Original work

Software Architecture

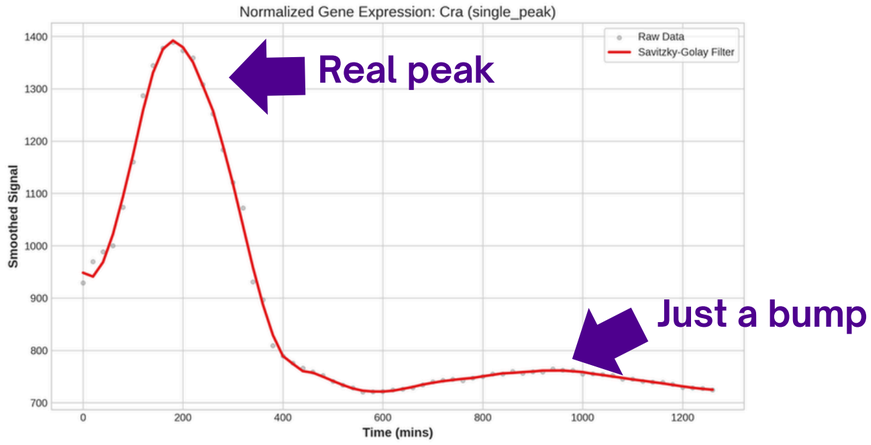
SAVITZKY-GOLAY FILTERING

- Smooths high-frequency noise using local low-degree polynomial regression
- poiynomial order = 2 or 3
- window size = 10-15% of length of dataset
- Durbin-Watson score is used to measure quality of smoothing (indirect)





PEAK/TROUGH DETECTION (main feature)



Source: Original work

Scipy.find_peaks is used BUT without static thresholds!

- Amplitude
  - Threshold = 10% of IQR (Inter Quartile Range)
  - Threshold = 10% * MAD * 1.4826
- Width
  - FWHM set based on time-units of experiment + time-scale of protein translation

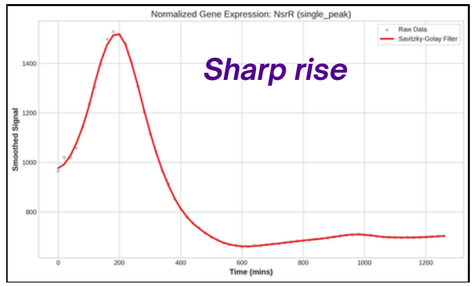
For TROUGH ... same thing but with the signal reversed

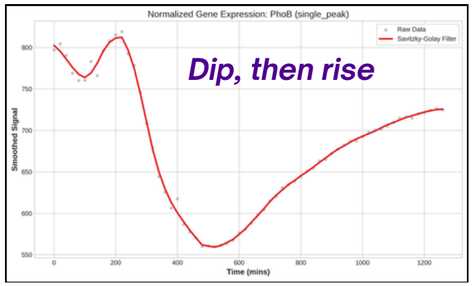
Classification - Decision Tree

DECISION TREE LOGIC for SHAPE CLASSIFICATION

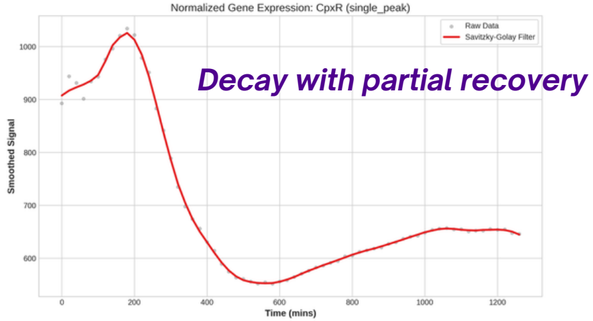
- Single peak
- Single Trough
- Sustained Activation
- Sustained Repression
  - Monotonic
  - Recovery
- Bi-phasic
- Repeated cycles
  - Oscillatory
  - Complex Bursting
- *Unclassfied*
- *Flat/No Response*

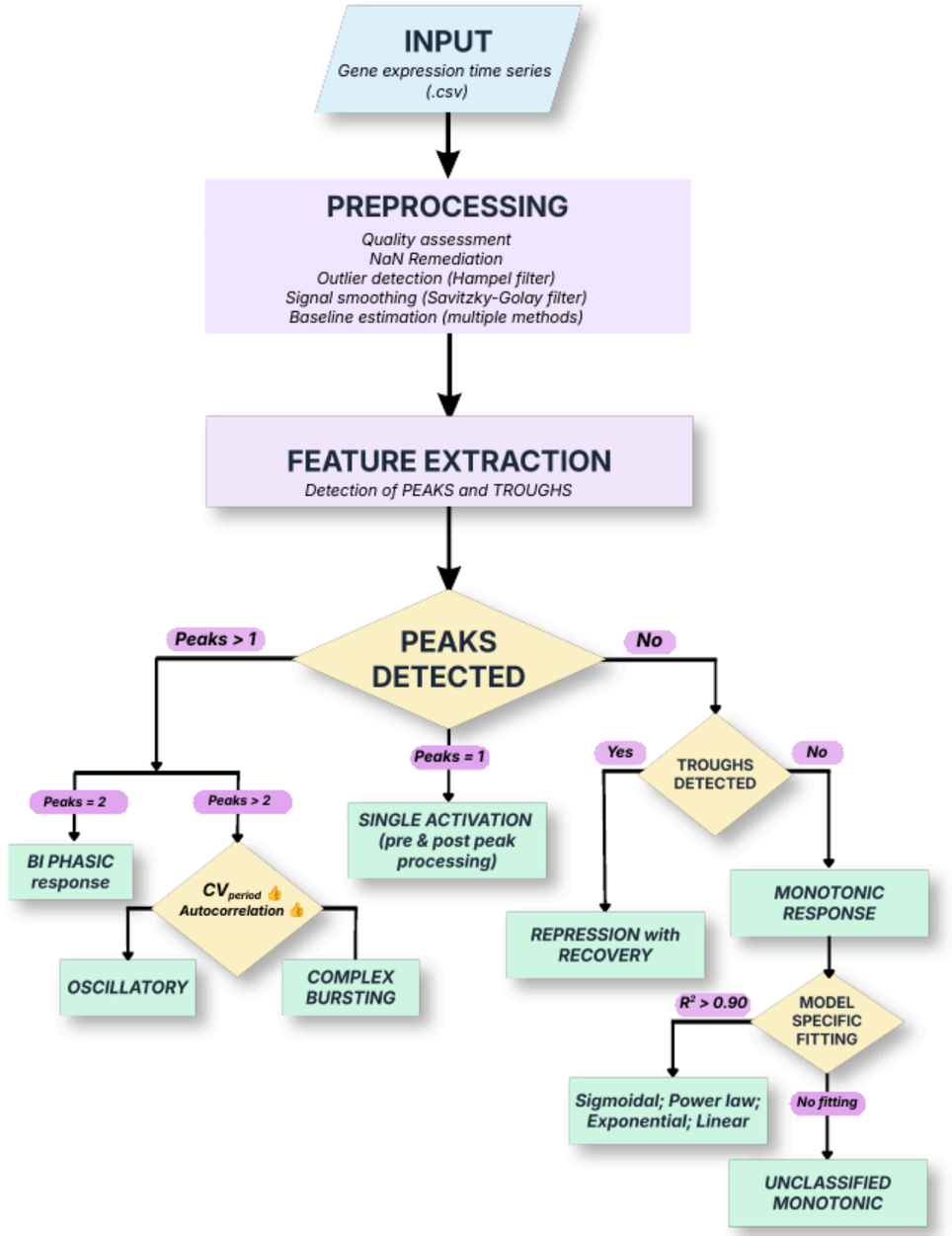
The tool ALSO classifies the PRE and POST peak events!











Metric Extraction

Common Metrics
- dw_score
- baseline
- category

Single Peak
- peak_time
- absolute_amplitude
- net_rise
- fold_change
- rise_time
- fall_time
- fwhm_width
- total_yield_auc
- degradation_rate
- half_life

Biphasic
- transition_delay
- net_amplitude_ratio
- phase_dominance_profile
- inter_phase_clearance_pct

Single Trough
- trough_time
- trough_value
- net_drop
- fold_change_down
- recovery_time
- recovery_pct

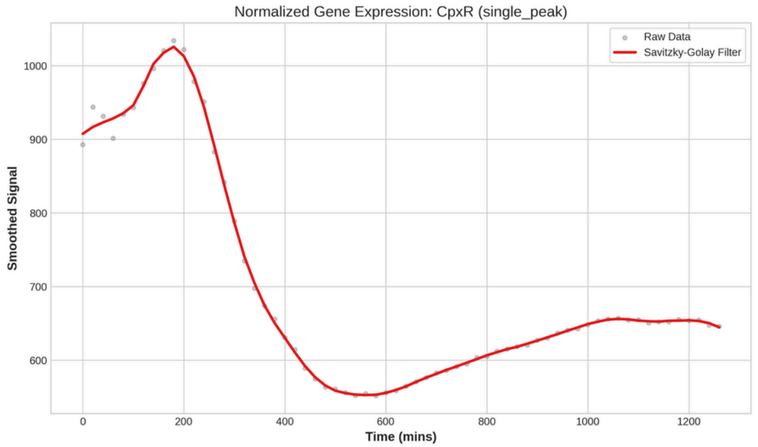
Oscillatory
- mean_period
- period_regularity_cv
- mean_pulse_height
- terminal_attenuation_ratio
- envelope_classification

Complex Irregular Bursting
- num_peaks
- mean_period
- cv_period

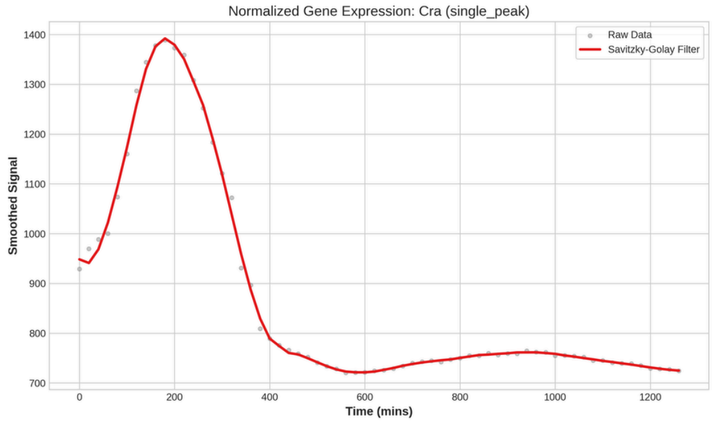
Monotonic Response
- direction
- total_change_absolute
- average_slope
- area_under_curve
- r_squared

Flat Response
- dynamic_range
- mean_signal_strength

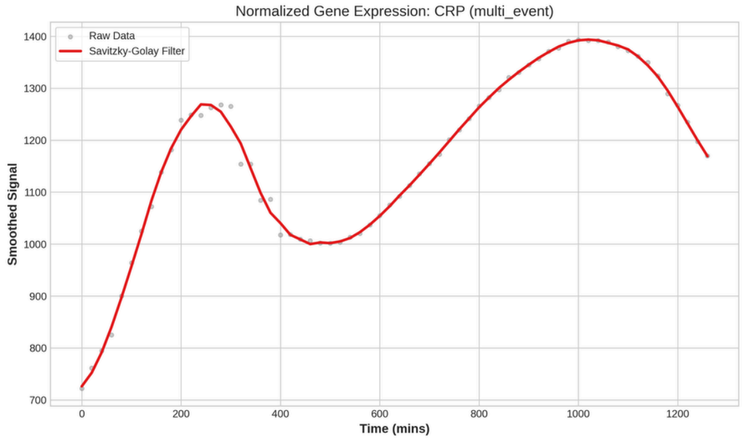
Validation of E. Coli. GLOBAL REGULATORS



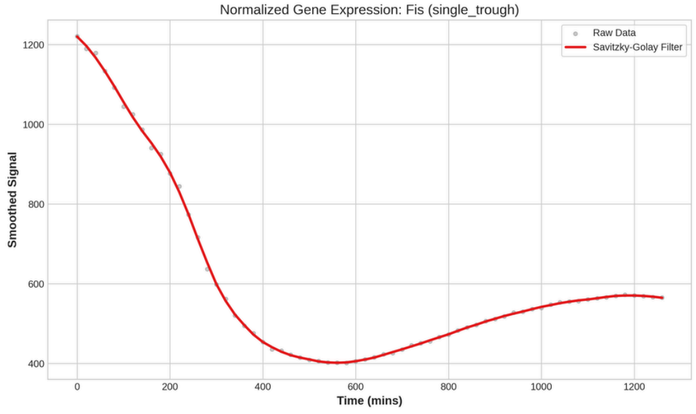
CpxR



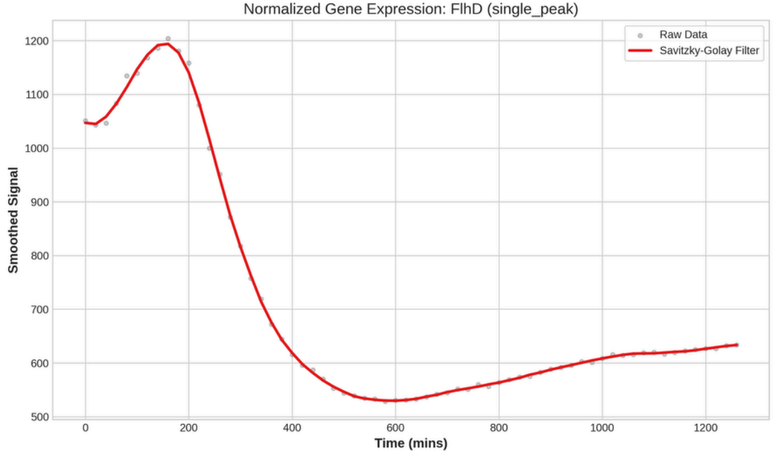
Cra



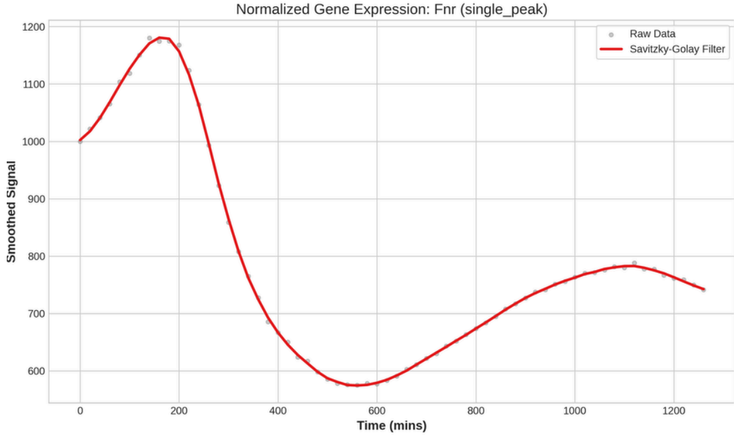
CRP



Fis

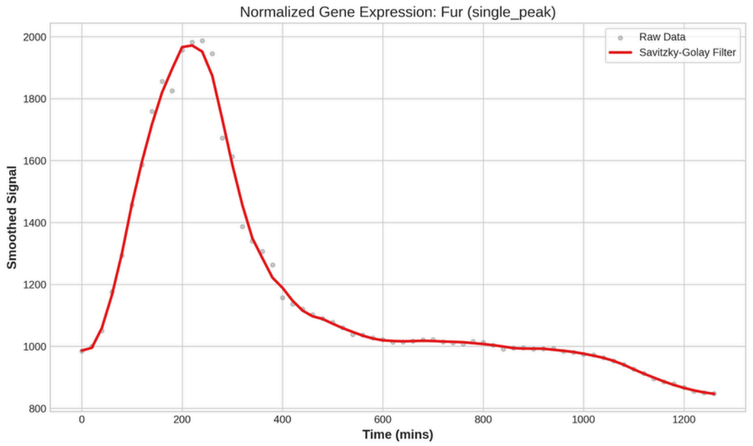


FlhD

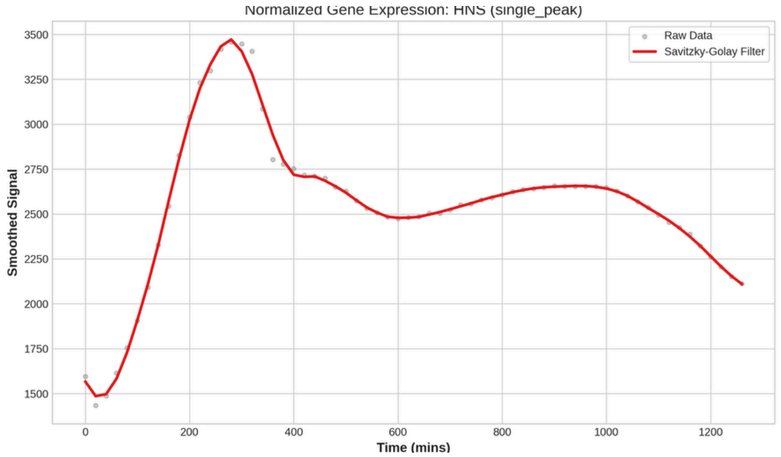


Fnr

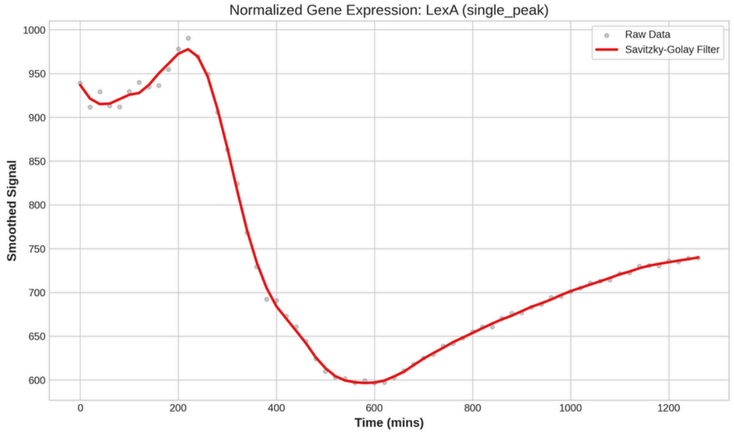
Validation of E. Coli. GLOBAL REGULATORS



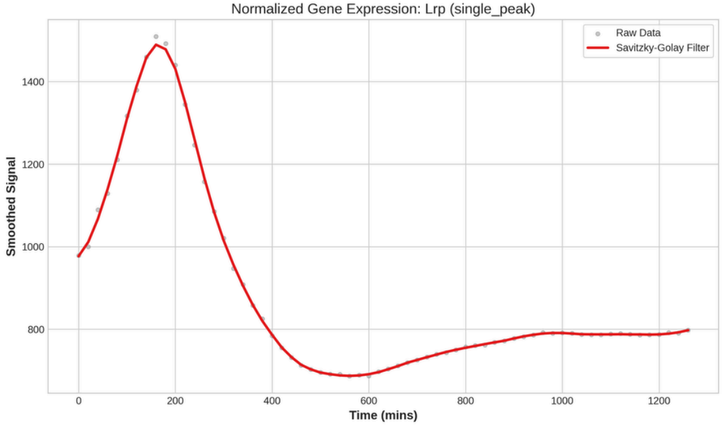
Fur



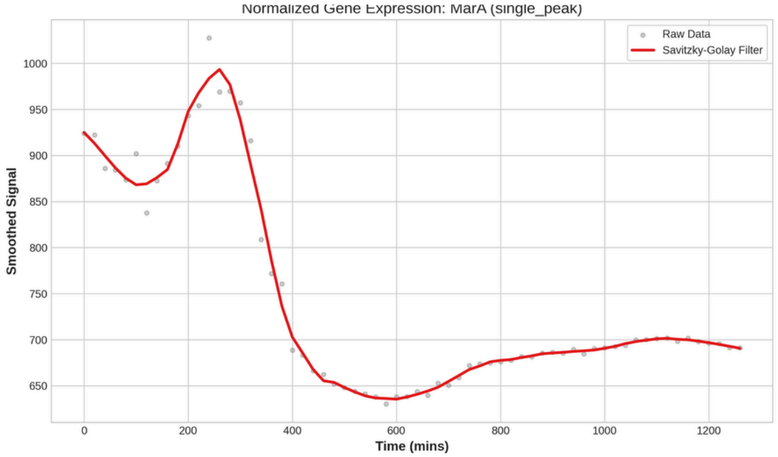
HNS



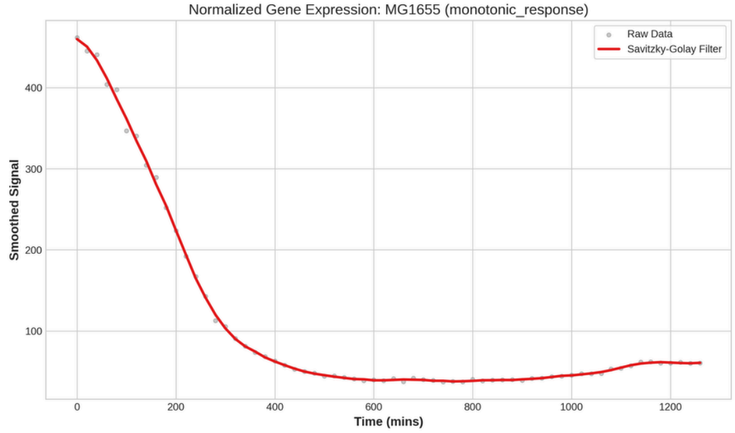
LexA



Lrp

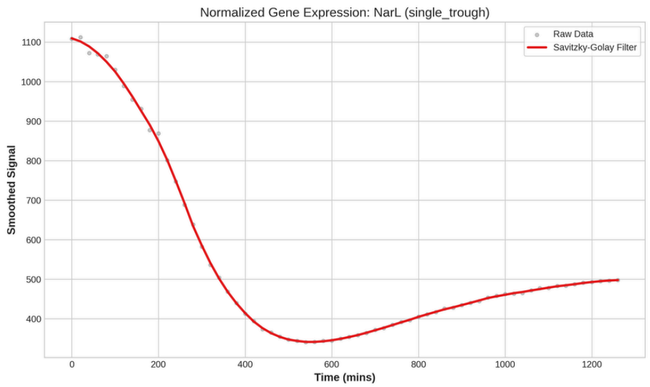


MarA

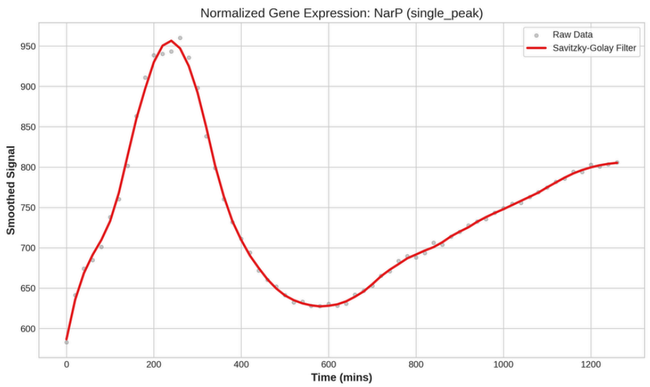


MG1655 (control)

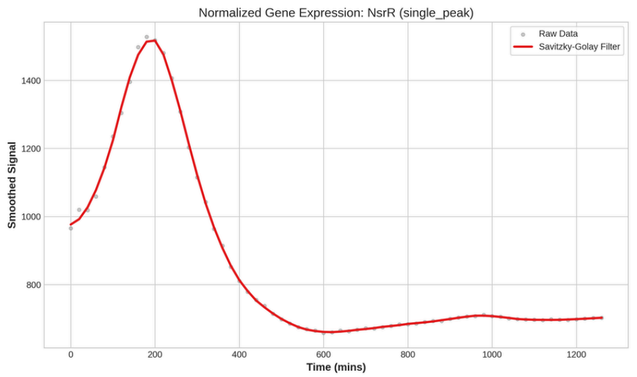
Validation of E. Coli. GLOBAL REGULATORS



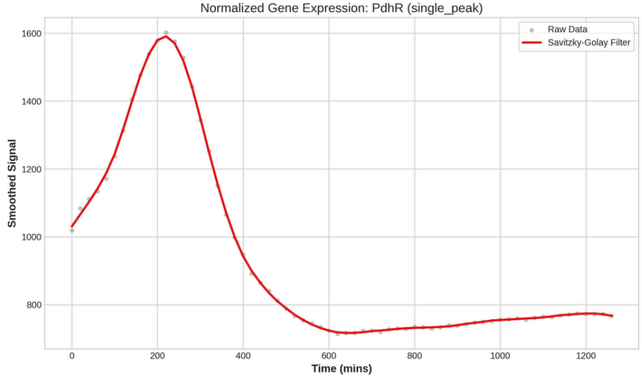
NarL



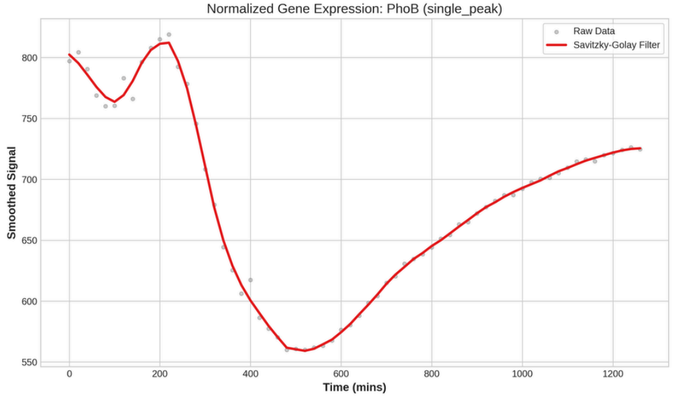
NarP



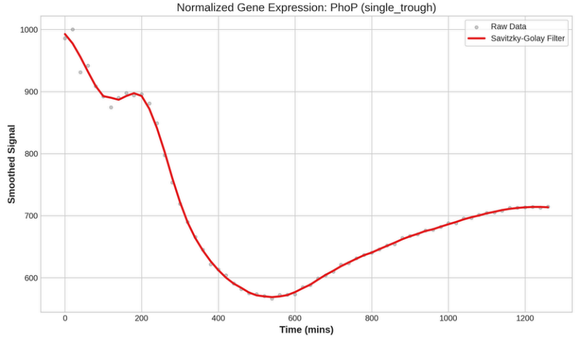
NsrR



PdhR

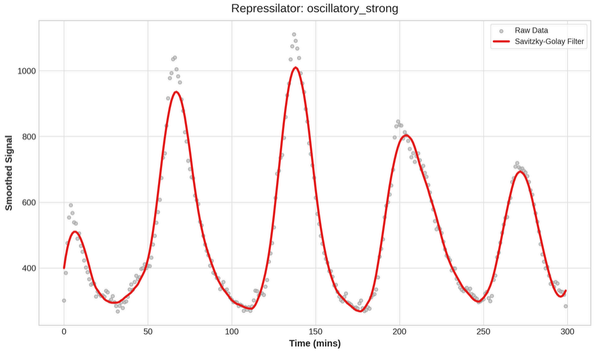


PhoB



PhoP

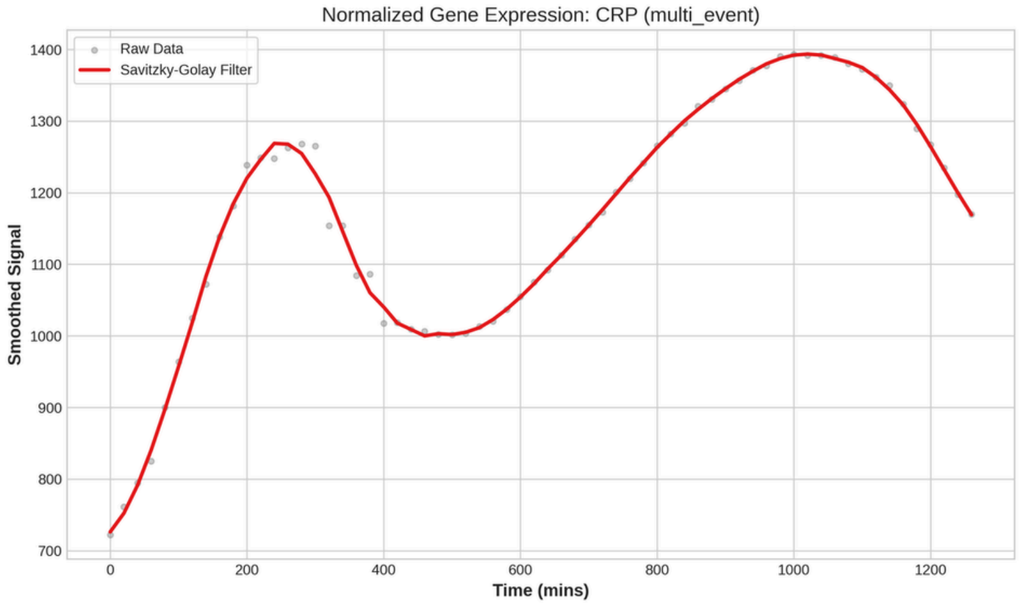
Oscillatory
(Simulation generated)



How to run the tool ??

The tool can run in the following modes:

- Sample mode
  - Replicates (confidence interval)
  - Average replicate response
- Batch mode
  - Multiple samples
  - Useful for comparison



<< Shape classification

Currenly only supports bash cmd
(Webserver coming soon ...)

```
gxa -i expression_data.csv -o ./results --plots
```

```
gxa --input data/replicates.csv \
    --output ./precision_analysis \
    --genes LexA recA \
    --baseline-method early_window \
    --prominence 0.15 \
    --prominence-mode statistical \
    --remove-outliers \
    --outlier-sensitivity 3.0 \
    --plots \
    --log-level DEBUG
```



Every threshold and decision-making parameter is tunable by the user

Extraction of metrics >>

```
CRP:
  dw_score: 2.820
  baseline: 726.156
  baseline_method: initial_value
  category: biphasic
  metadata:
    num_peaks: 2
    metrics:
      transition_delay: 780.000
      net_amplitude_ratio: 4.410
      phase_duration_ratio: 1.739
      composite_symmetry_score: 0.401
      phase_dominance_profile: secondary_phase
      inter_phase_clearance_pct: 178.800
      amplitude_symmetry_component: 0.227
      temporal_symmetry_component: 0.575
```

Acknowledgements



RESOURCES BY

Laboratory of Biosystem Dynamics

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Tampere University

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INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
भारतीय प्रौद्योगिकी संस्थान खड़गपुर
1951
योगः कर्मसु कौशलम्

75
राष्ट्र सेवार्थ समर्पित
1951
DEDICATED TO THE SERVICE OF THE NATION
IIT Kharagpur
PLATINUM JUBILEE

Questions??

